

Graphing Basic Polar Curves from Equations

Answer Key

Precalculus

Quick Answers

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|---|--|
| 1. 6 | 8. (a) r at $0 = 6$, (b) r at $\pi = -2$ |
| 2. 3 | 9. 4.24 |
| 3. $0, \frac{\pi}{3}, \frac{2\pi}{3}$ | 10. (a) r at $3\pi/2 = -2$, — |
| 4. (a) r at $3\pi/2 = 8$, — | 11. (a) angles where the two r -values agree = $\frac{\pi}{4}$, |
| 5. $\frac{1}{2}$ | $\frac{5\pi}{4}$, — |
| 6. (a) r at $\pi/6 = \frac{3}{2}$, — | 12. (a) the positive $r = 4$, (a) the matching |
| 7. $0, \pi$ | angle = $\frac{7\pi}{6}$, — |
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What is verified

14 of 19 answers machine-checked against SymPy, across 12 problems.

— marks an answer only you can judge — problems 4, 6, 10, 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus

HSF-IF.C.7 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–5 of 5 across 19 tagged checks.

Common wrong answers

1. If they got 7.46: used sine instead of cosine when substituting
 5. If they got 3.5: took $\cos(2\pi/3)$ as $+1/2$ instead of $-1/2$
 9. If they got 2.3: forgot to double the angle before taking the sine
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1. The cardioid $r = 4 + 4 \cos \theta$ is graphed one point at a time. Find the exact value of r when $\theta = \pi/3$.

Solution: $\cos \frac{\pi}{3} = \frac{1}{2}$, so $r = 4 + 4 \left(\frac{1}{2} \right) = 4 + 2$.

$r = 6$

2. The graph of $r = 6 \cos \theta$ is a circle that sits on the polar axis. Find r when $\theta = \pi/3$.

Solution: $r = 6 \cos \frac{\pi}{3} = 6 \left(\frac{1}{2} \right)$, and the point $(3, \pi/3)$ lies on the circle of diameter

6.

$r = 3$

3. Find every θ in $[0, \pi)$ for which $r = 5 \sin(3\theta)$ equals zero.

Solution: $5 \sin(3\theta) = 0$ exactly when $\sin(3\theta) = 0$, i.e. $3\theta = 0, \pi, 2\pi, \dots$, so $\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \dots$. Only the first three of these lie in $[0, \pi)$. These are the angles at which the three-petal rose returns to the pole.

$$\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}$$

4. Cardioid $r = 4 - 4 \sin \theta$. (a) Find r at $\theta = 3\pi/2$. (b) Explain why the curve touches the pole at exactly one angle in one full turn.

Solution (a): $\sin \frac{3\pi}{2} = -1$, so $r = 4 - 4(-1) = 4 + 4$.

$$r = 8$$

Solution (b) — instructor-judged. $r = 0$ requires $4 - 4 \sin \theta = 0$, that is $\sin \theta = 1$. On $[0, 2\pi)$ the sine equals 1 at the single angle $\theta = \pi/2$, so the cardioid arrives at the pole once and leaves again rather than passing through it twice. Full credit for a response that reduces to $\sin \theta = 1$ and notes that it has exactly one solution per turn.

5. Find the exact value of r for the limaçon $r = 2 + 3 \cos \theta$ at $\theta = 2\pi/3$.

Solution: $\cos \frac{2\pi}{3} = -\frac{1}{2}$ (second quadrant, so the cosine is negative), giving $r = 2 + 3 \left(-\frac{1}{2}\right) = 2 - \frac{3}{2}$.

$$r = \frac{1}{2}$$

6. Rose $r = 3 \cos(2\theta)$. (a) Find r at $\theta = \pi/6$. (b) Sketch the whole curve and label the petal tip on the polar axis.

Solution (a): $2\theta = \frac{\pi}{3}$ and $\cos \frac{\pi}{3} = \frac{1}{2}$, so $r = 3 \left(\frac{1}{2}\right)$.

$$r = \frac{3}{2}$$

Solution (b) — instructor-judged. Because $n = 2$ is even, the rose has $2n = 4$ petals, each of length 3. One petal points along the polar axis with its tip at a distance of 3 from the pole; the other three tips lie a quarter turn apart. The curve returns to the pole at $\theta = \pi/4, 3\pi/4, 5\pi/4, 7\pi/4$. Award full credit for a freehand four-petal sketch whose axis petal is labelled with the distance 3.

7. The circle $r = 8 \sin \theta$ passes through the pole. Find every θ in $[0, 2\pi)$ for which $r = 0$.

Solution: $8 \sin \theta = 0$ when $\sin \theta = 0$, which on $[0, 2\pi)$ happens at $\theta = 0$ and $\theta = \pi$. Both angles give the same point — the pole — which is why the circle is traced completely as θ runs over only half a turn.

$$\theta = 0 \text{ and } \theta = \pi$$

8. Limaçon $r = 2 + 4 \cos \theta$. (a) Find r at $\theta = 0$. (b) Find r at $\theta = \pi$.

Solution (a): $\cos 0 = 1$, so $r = 2 + 4(1)$.

$$r = 6$$

Solution (b): $\cos \pi = -1$, so $r = 2 + 4(-1) = 2 - 4$. Because $a = 2 < b = 4$, this limaçon has an inner loop and its r -values go negative.

$$r = -2$$

9. For the rose $r = 6 \sin(2\theta)$, find r at $\theta = \pi/8$, rounded to two decimal places.

Solution: Double the angle first: $2\theta = \frac{\pi}{4}$. Then $r = 6 \sin \frac{\pi}{4} = 6 \left(\frac{\sqrt{2}}{2} \right) = 3\sqrt{2} = 4.2426 \dots$

$$r \approx 4.24$$

10. Limaçon $r = 1 + 3 \sin \theta$. (a) Find r at $\theta = 3\pi/2$. (b) Describe where that point lands and what the negative r -values trace.

Solution (a): $\sin \frac{3\pi}{2} = -1$, so $r = 1 + 3(-1) = 1 - 3$.

$$r = -2$$

Solution (b) — instructor-judged. A negative r reverses the ray: from the pole, face $3\pi/2$ (straight down) and walk 2 units backwards, which lands 2 units up, at the point $(2, \pi/2)$. The angles where $1 + 3 \sin \theta < 0$ are exactly the ones that draw the small inner loop. Full credit requires both the reversed direction and the identification of the inner loop.

11. The circles $r = 6 \cos \theta$ and $r = 6 \sin \theta$ cross twice. (a) Solve $6 \cos \theta - 6 \sin \theta = 0$ on $[0, 2\pi)$. (b) Sketch both circles and mark the crossing point that is not the pole.

Solution (a): $6 \cos \theta = 6 \sin \theta \Rightarrow \tan \theta = 1$, so $\theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$.

$$\theta = \frac{\pi}{4} \text{ or } \theta = \frac{5\pi}{4}$$

Solution (b) — instructor-judged. $r = 6 \cos \theta$ is the circle of diameter 6 resting on the polar axis; $r = 6 \sin \theta$ is the same circle rotated onto the vertical axis. They meet at the pole and again on the ray $\theta = \pi/4$, where both equations give $r = 3\sqrt{2}$. Award full credit for two correctly placed circles with the non-pole crossing marked on the $\pi/4$ ray.

12. A point is given as $(-4, \pi/6)$. (a) Rewrite it with a positive first coordinate and an angle in $[0, 2\pi)$. (b) Explain why both pairs mark the same location.

Solution (a): Reversing the ray adds half a turn and makes the distance positive: the distance is $|-4|$ and the angle is $\frac{\pi}{6} + \pi = \frac{7\pi}{6}$.

$$r = 4 \quad \text{and} \quad \theta = \frac{7\pi}{6}$$

Solution (b) — instructor-judged. A negative first coordinate means "walk backwards along the ray". Walking 4 units backwards along the $\pi/6$ ray is the same physical motion as walking 4 units forwards along the opposite ray, $\theta = 7\pi/6$. Full credit for any response that names the reversal of the ray as the reason.